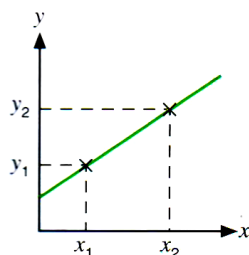


$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

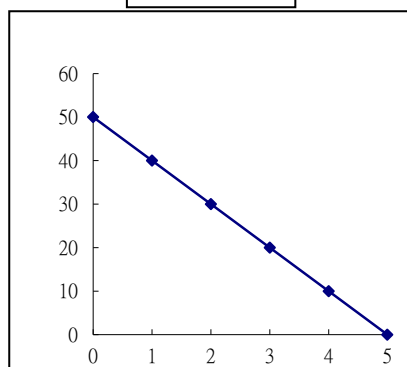
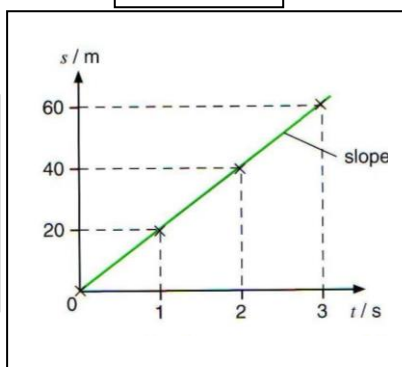


$$\text{slope} = \frac{\text{Rise}}{\text{Run}}$$

+ slope

- slope

Slope =

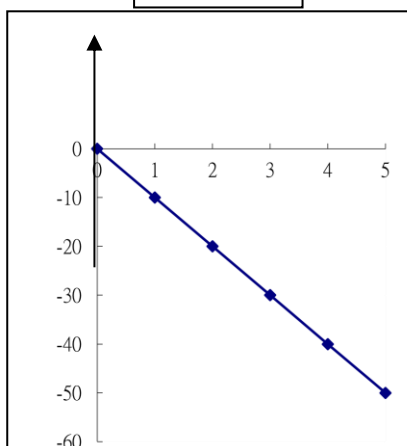
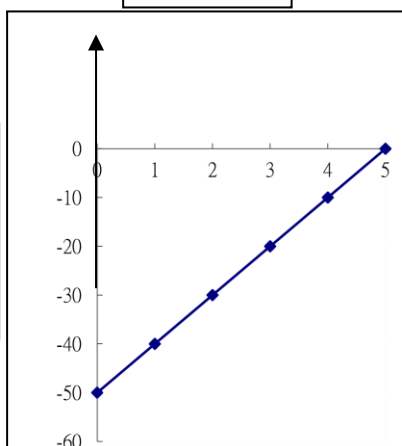


Slope =

+ slope

- slope

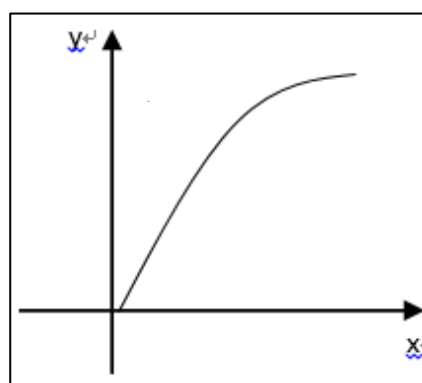
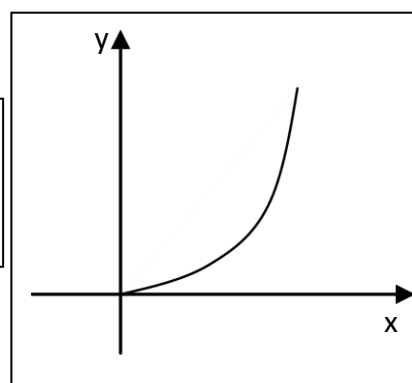
Slope =



Slope =

Uniform, Increasing, Decreasing

Slope



s-t

v-t

a-t

曲

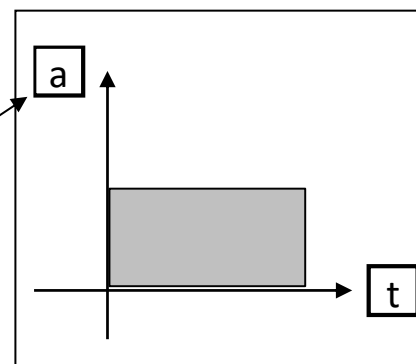
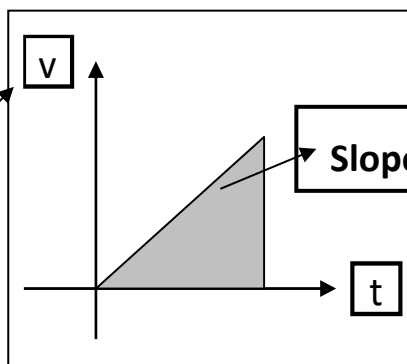
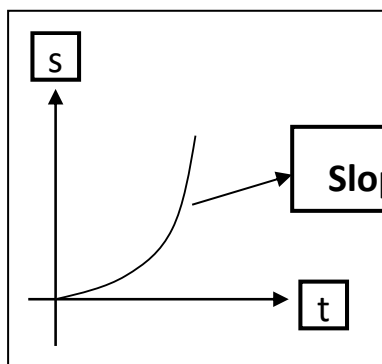
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無

Slope of the s-t graph is equal to **velocity (v)**

Slope of the v-t graph is equal to **acceleration (a)**

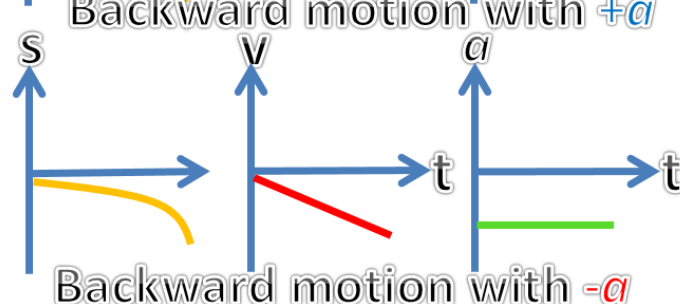
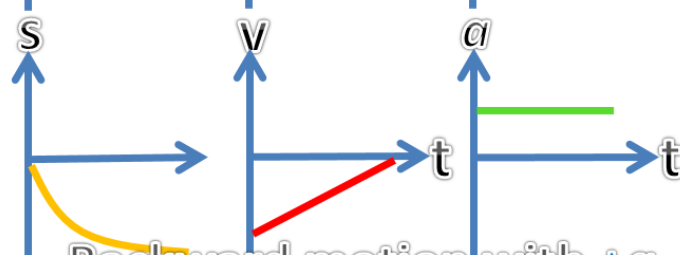
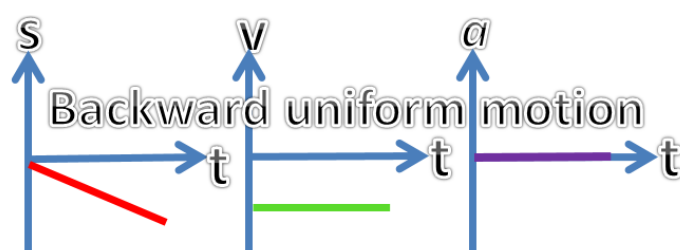
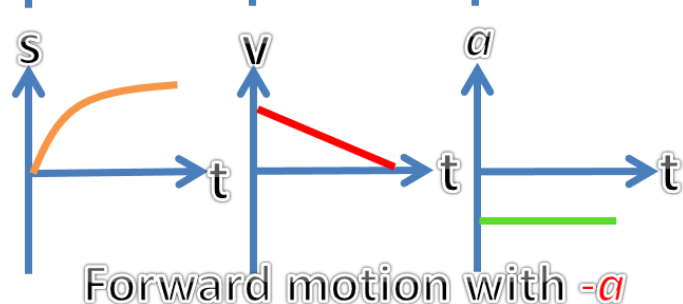
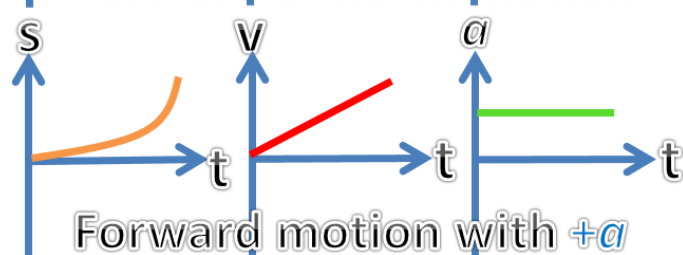
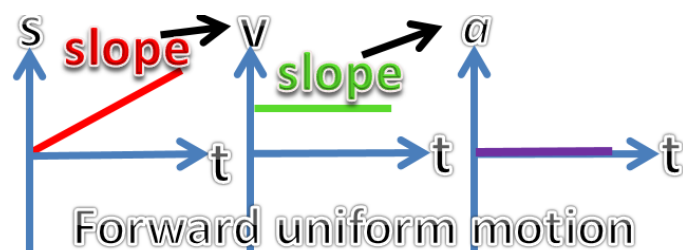


The **area** under v-t graph is equal to **total displacement (s)**

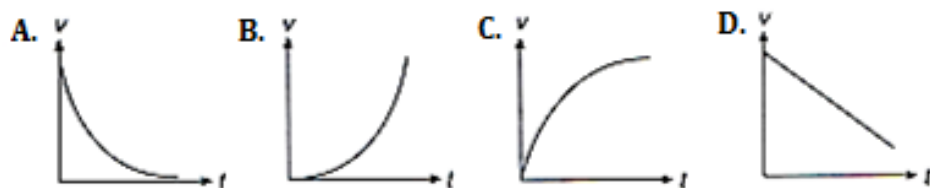
Area under a-t graph is equal to **change of velocity (v)**

Remarks:

- (1) if a body travel along a straight line without returning back, the displacement-time (s-t) graph is **identical to** the distance-time graph.
- (2) From v-t graph: A **steeper slope** represents an **acceleration with a larger magnitude**. The object **changes its direction** of motion when the **sign** of its **velocity changes** (when the graph crosses the horizontal axis).

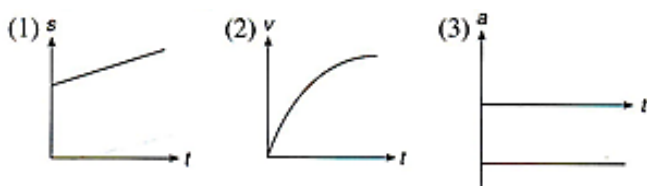


1. Which of the following $v - t$ graphs shows a uniformly decelerating motion?



D

2. Which of the following graphs represent(s) uniformly accelerating motion?

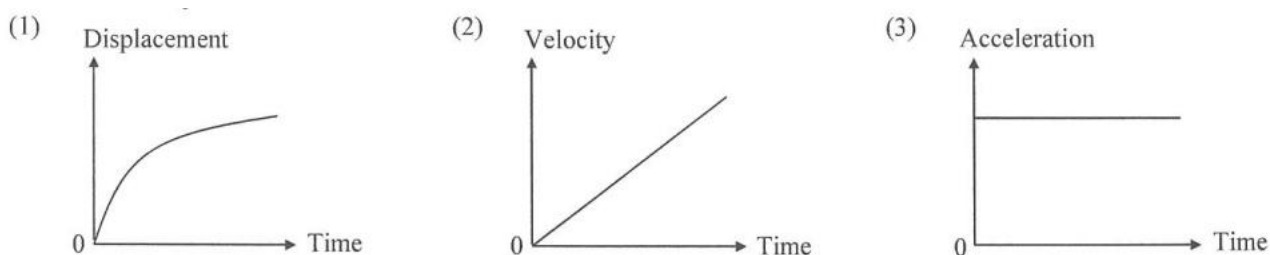


- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

B

3. 1996/II/4

An object moves with uniform acceleration along a straight line. Which of the following graphs correctly

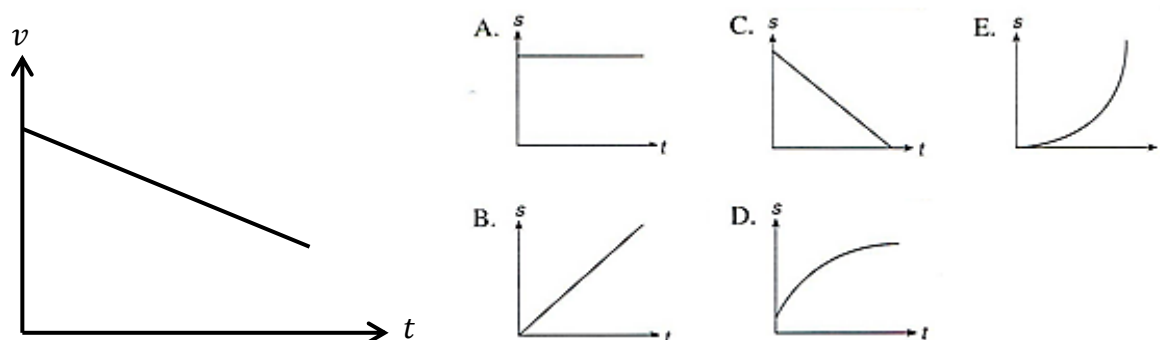


- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

- × (1) Slope of $s \sim t$ graph = velocity.
Decreasing slope $\Rightarrow$ decreasing velocity.
 $\therefore$ The $s \sim t$ graph with decreasing slope represents uniform deceleration.
- ✓ (2) Slope of $v \sim t$ graph = acceleration.
Slope of $v \sim t$ graph is positive and is a straight line $\Rightarrow$ a uniform acceleration.
- ✓ (3) Constant positive acceleration $\Rightarrow$ uniform acceleration

D

4. Which of the following graphs correctly shows the variation of distance travelled by the object?



D

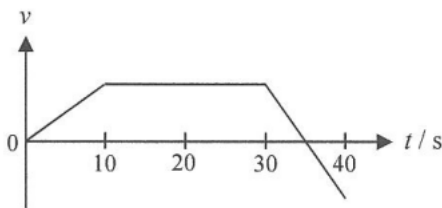
5. 1998/II/4

slope of $v \sim t$ graph = a

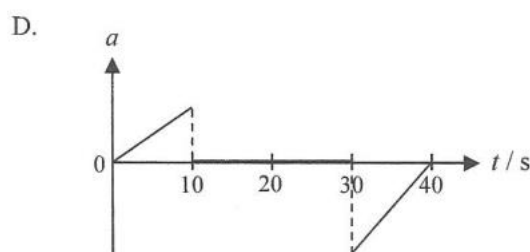
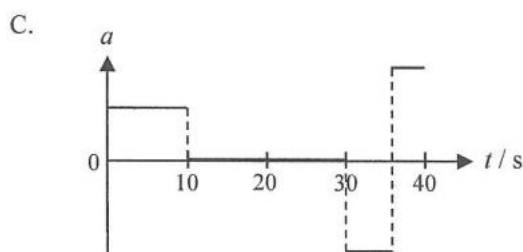
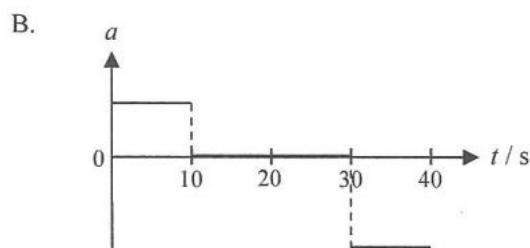
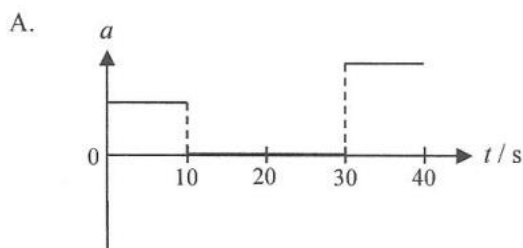
For $t = 0$ s to $t = 10$ s, slope of $v \sim t$ graph is (+) $\Rightarrow a$ is (+)

For $t = 10$ s to $t = 30$ s, slope of $v \sim t$ graph is 0 $\Rightarrow a = 0$

For $t = 30$ s to $t = 40$ s, slope of $v \sim t$ graph is (-) $\Rightarrow a$ is (-)

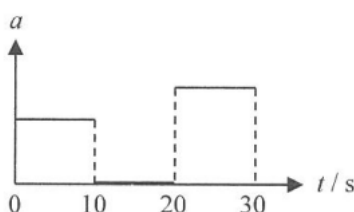


The velocity-time graph of a car travelling along a straight horizontal road is shown above. Which of the following graphs shows the variation of the acceleration a of the car with time t ?

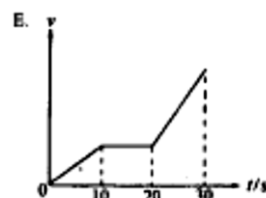
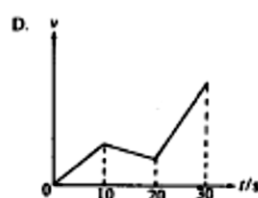
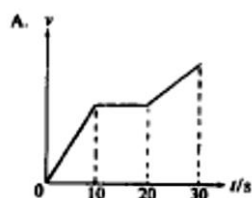


B

6. 1999/II/3



A car starts from rest and travels along a straight road. The acceleration-time graph of the car is shown above. Which of the following graphs shows the variation of the velocity v of the car with time t ?



E

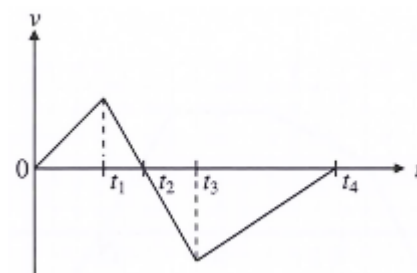
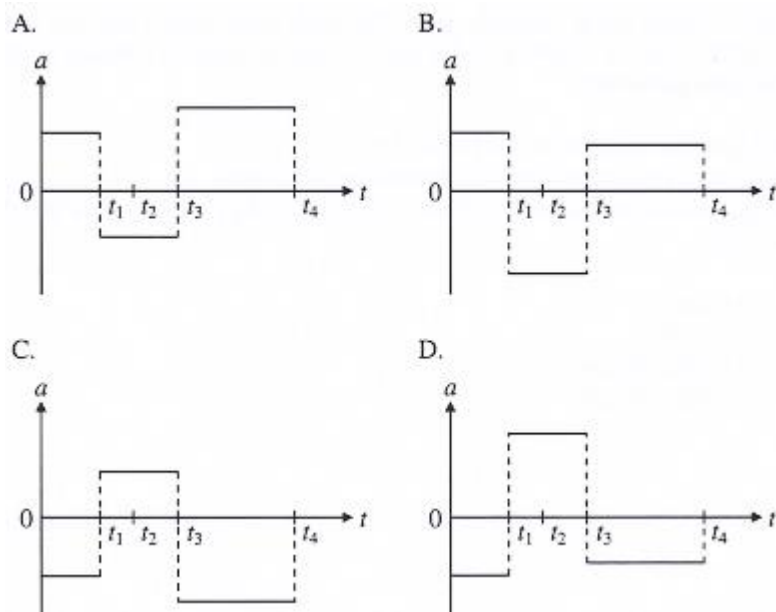
As slope of $v \sim t$ graph = acceleration,

When $t = 0$ s to $t = 10$ s, $a : (+) \Rightarrow$ slope of $v \sim t$ graph : (+)

When $t = 10$ s to $t = 20$ s, $a : (+) \Rightarrow$ slope of $v \sim t$ graph : 0 (a horizontal line)

When $t = 20$ s to $t = 30$ s, $a : (+)$ with larger value $\Rightarrow$ slope of $v \sim t$ graph : (+) with larger slope

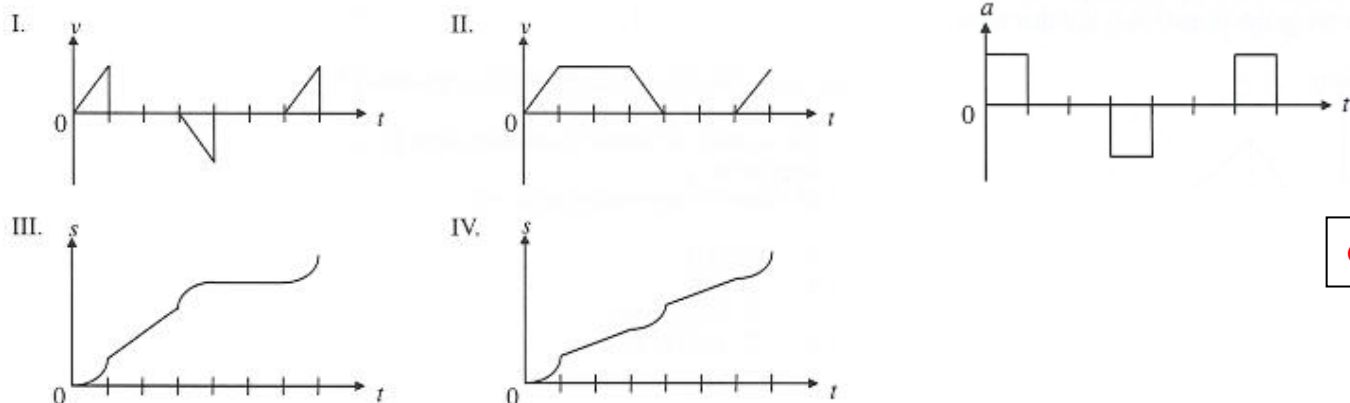
7. The graph above shows the velocity-time graph of an object moving along a straight line. Which of the following graphs correctly sketches its acceleration-time graph?



B

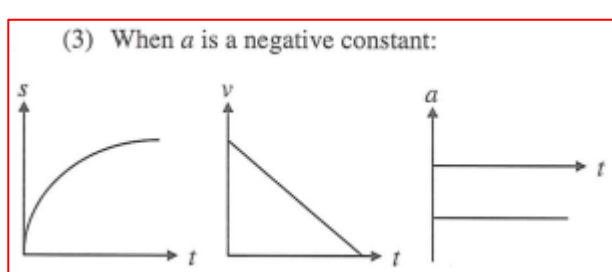
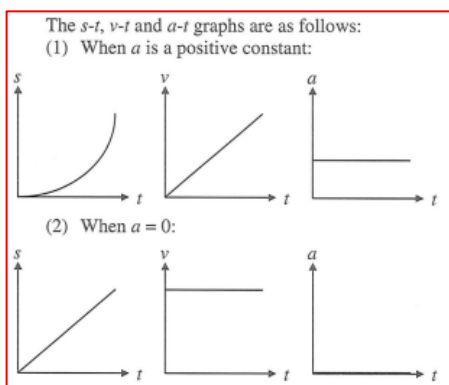
Slope of $v-t$ graph = acceleration
Consider the $v-t$ graph
Slope from $t = 0$ to $t = t_1$ and slope from $t = t_3$ to $t = t_4$ are positive.
Slope from $t = t_1$ to $t = t_3$ is negative.
Also slope from $t = 0$ to $t = t_1 >$ slope from $t = t_3$ to $t = t_4$, so acceleration from $t = 0$ to $t = t_1 >$ acceleration from $t = t_3$ to $t = t_4$.

8. The figure above shows the acceleration-time graph of a body. Which of the following graphs are the corresponding velocity-time graph and displacement-time graph?



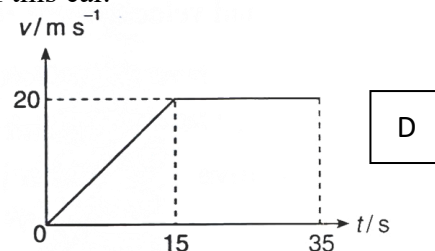
C

- A. I and III
B. I and IV
C. II and III
D. II and IV



9. The $v-t$ graph of a moving car is shown. Find the total displacement of this car.

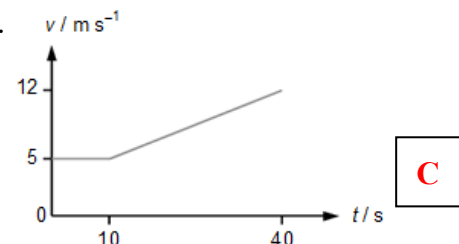
A. 250 m
B. 300 m
C. 350 m
D. 550 m



D

10. John is driving along a straight road at a uniform velocity. At time $t = 10$ s, he steps on the accelerator pedal and his car travels with a uniform acceleration. The graph below shows how the speed of the car v varies with time t . Find the distance travelled by the car from time $t = 0$ to 40 s.

A. 240 m
B. 290 m
C. 305 m
D. 480 m



C

11. 1995/II/8

The above diagram shows the variations of the velocity of an object with time. What is the distance travelled by the object in the first 3 seconds?

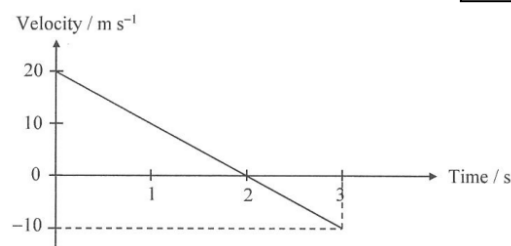
A. 5 m
B. 15 m
C. 25 m
D. 30 m

Distance travelled = total area of the graph between the line and the x-axis

As distance is a scalar, the direction is not relevant.

Thus, the absolute value of the area represents the distance travelled.

$$\therefore s = \frac{1}{2}(20)(2) + \frac{1}{2}(3-2)(10) = 25 \text{ m}$$



C

12. A bead is projected up along a smooth incline at point P at time $t = 0$. After reaching the highest point, the bead then travels downwards and passes point Q at $t = 3$ s as shown. The velocity-time ($v-t$) relationship of the bead is shown in the graph below. Find the separation PQ along the incline.

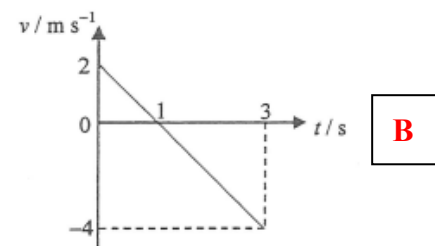
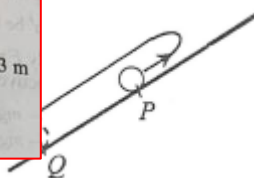
A. 2 m
B. 3 m
C. 4 m
D. 5 m

Method 1

Note that displacement is area under $v-t$ graph.

$$\text{The displacement of the bead} = \frac{(1-0)(2-0)}{2} - \frac{(3-1)[0-(-4)]}{2} = -3 \text{ m}$$

$\therefore$ The separation PQ along the incline is 3 m.



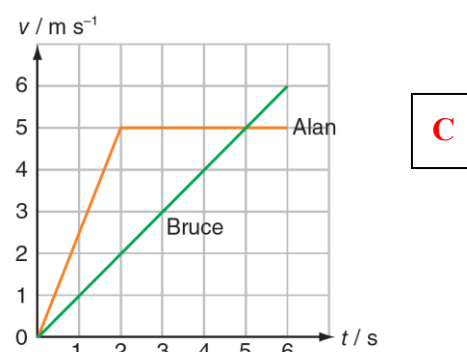
B

13. Alan and Bruce participate in a 60-m race along a straight track. They start from the same position and their $v-t$ graphs for the first 6 s are as shown. What is the maximum distance between Alan and Bruce in the first 6 s?

A. 3 m
B. 5 m
C. 7.5 m
D. 12.5 m

Alan is ahead of Bruce and they become further and further apart when Alan's speed is higher than Bruce's.
 $\therefore$ Distance between them is the largest at $t = 5$ s

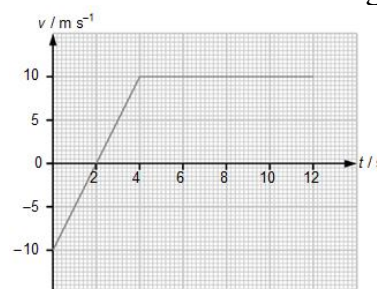
$$\begin{aligned} \text{Maximum distance} &= \text{distance travelled by Alan} \\ &\quad - \text{distance travelled by Bruce} \\ &= \frac{(3+5) \times 5}{2} - \frac{5 \times 5}{2} \\ &= 7.5 \text{ m} \end{aligned}$$



C

14. The velocity–time (v – t) graph of a car travelling on a straight road is shown below. What is the average velocity of the car from $t = 0$ to $t = 12$ s?

- A. 5.83 m s^{-1}
B. 6.67 m s^{-1}
C. 7.5 m s^{-1}
D. 8.33 m s^{-1}



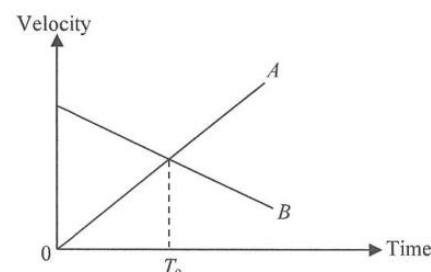
B

15. 2000/II/4

Two cars A and B travel along the same straight road. The velocity-time graphs of the two cars are shown above. Which of the following statements about the motion of the two cars at time T_0 is/are always correct?

- (1) The two cars meet each other
(2) The two cars are moving with the same speed but in opposite directions
(3) Car A is accelerating while B is decelerating

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only



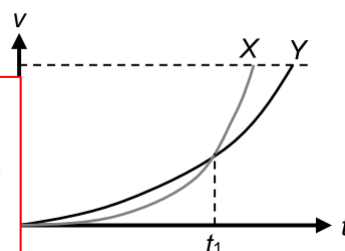
B

16. Two cars X and Y are racing on a straight road. The figure shows their velocity–time graphs from the beginning of the race. They are at the same position at time $t = 0$. Which of the following statements is/are correct?

- (1) X overtakes Y at $t = t_1$.
(2) Y has a higher average velocity than X from $t = 0$ to $t = t_1$.
(3) X has a higher instantaneous acceleration than Y at $t = t_1$.

- A. (1) only
B. (3) only
C. (1) and (2)
D. (2) and (3)

From $t = 0$ to $t = t_1$, the area under the v – t graph of X is smaller than that of Y. That means X has a smaller displacement than Y and does not overtake Y.
Since the displacement of Y from $t = 0$ to $t = t_1$ is larger than that of X, Y has a higher average velocity.
The instantaneous acceleration can be determined by the slope of the tangent of the v – t graph. At $t = t_1$, the graph of X has a larger slope.



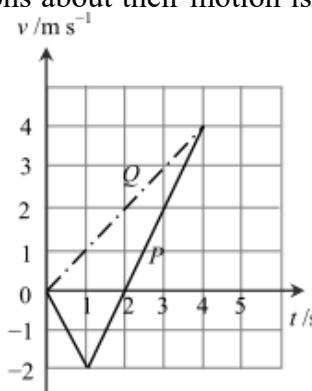
D

17. Two particles P and Q start from the same position and travel along the same straight line. The figure shows the velocity-time (v – t) graph for P and Q. Which of the following descriptions about their motion is/ are correct?

- (1) At $t = 1$ s, P changes its direction of motion.
(2) At $t = 2$ s, the separation between P and Q is 4 m
(3) At $t = 4$ s, P and Q meet each other

- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(1) ✗ The velocity of P remains negative when time passes $t = 1$ s.
(2) ✓ Note that the separation between P and Q is the difference in their displacements, which is the area between their v – t graphs.
 $\therefore$ At $t = 2$ s, the separation between P and Q $= \frac{1}{2}(2)(2) + \frac{1}{2}(2)(2) = 4$ m
At $t = 4$ s, the separation between P and Q $= \frac{1}{2}(2)(4) + \frac{1}{2}(2)(2) = 6$ m
(3) ✗ (See the explanation in (2).)



B

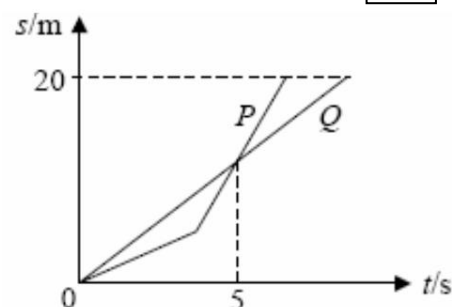
18. 2002/II/1

The figure above shows the distance-time graphs of two toy cars P and Q moving along a linear track. Which of the following statements are correct?

A

- (1) Car P will reach the 20 m-mark first.
- (2) Car P is overtaking car Q at $t = 5$ s.
- (3) The average speed of car P in the first 5 s is smaller than that of car Q .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)



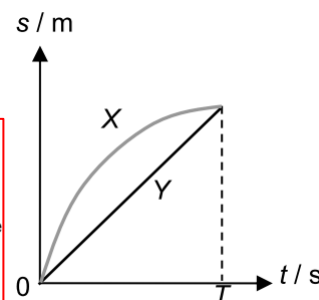
19. The figure below shows the $s-t$ graphs of objects X and Y which move along the same straight line. Which of the following statements is/are correct?

D

- (1) X keeps slowing down.
- (2) X and Y travel the same distance within the period shown.
- (3) The velocity of Y is greater than that of X at time T .

- A (1) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

The slope of $s-t$ graph is velocity. From the figure, X moves in positive direction and its velocity decrease with time.
 $\therefore$ (1) is correct.
 From the figure, X and Y always move in the positive direction and they have the same displacement within the period shown. Therefore, they travel the same distance within the period shown.
 $\therefore$ (2) is correct.
 The slope of $s-t$ graph of Y is larger than that of X at time T .
 $\therefore$ (3) is correct.



20. A toy car travels with constant acceleration along a straight line for a constant displacement. The graph above shows the variation of the square of its final velocity with the square of its initial velocity. If the acceleration of the car is 0.25 ms^{-2} , find the displacement of the toy car.

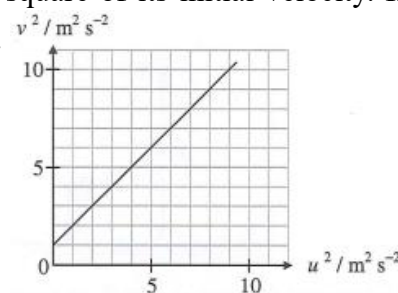
- A. 1 m
- B. 2 m
- C. 3 m
- D. 4 m

$$v^2 = u^2 + 2as$$

$$\text{y-intercept} = 2as = 1$$

$$2(0.25)s = 1$$

$$s = 2 \text{ m}$$



B

21. A truck of mass 2500 kg starts from rest and moves along a straight line. The graph above shows the variation of the square of time taken with the displacement of the truck. Find the acceleration of the truck.

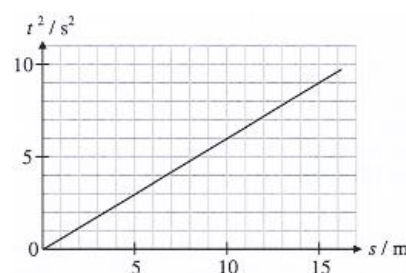
- A. 0.6 ms^{-2}
- B. 1.67 ms^{-2}
- C. 1.33 ms^{-2}
- D. 3.33 ms^{-2}

$$s = \frac{1}{2}at^2$$

$$t^2 = \frac{2s}{a}$$

$$\text{Slope} = \frac{2}{a} = \frac{3}{5}$$

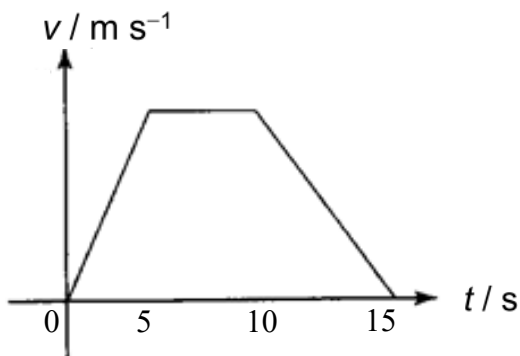
$$\therefore a = 3.33 \text{ m s}^{-2}$$



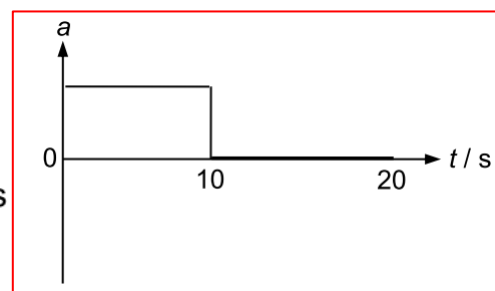
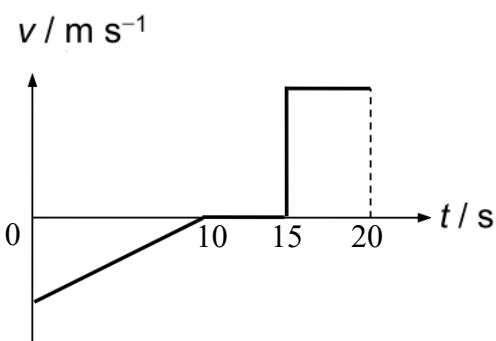
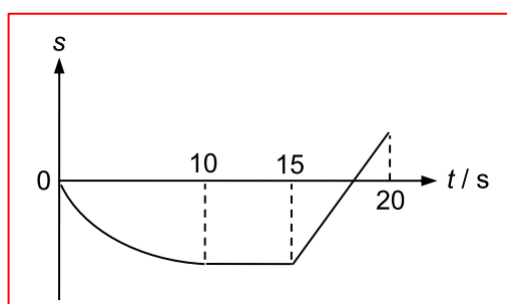
D

22. Describe the motions of the particle represented by the velocity-time graphs shown below and sketch their corresponding $s-t$ and $a-t$ graphs. (Assume all objects start at $s = 0$ when $t = 0$)

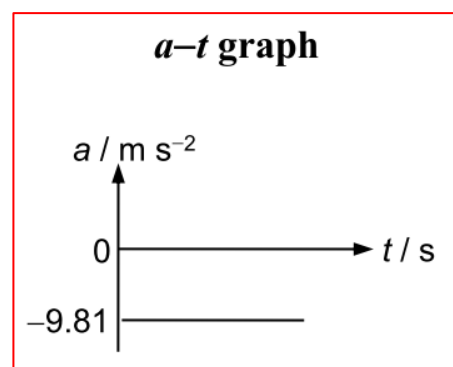
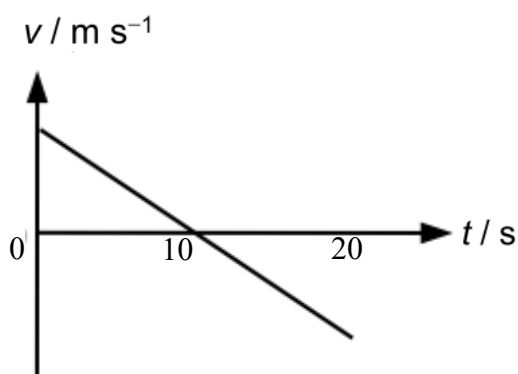
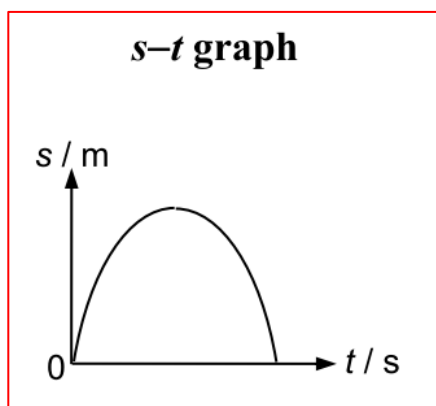
(a)



(b)

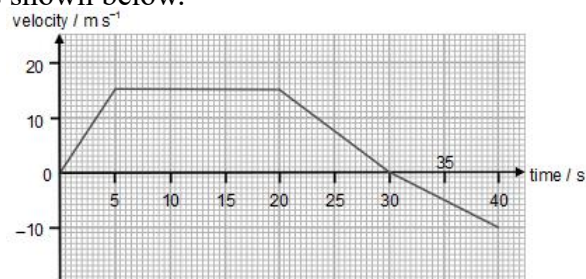


(c)

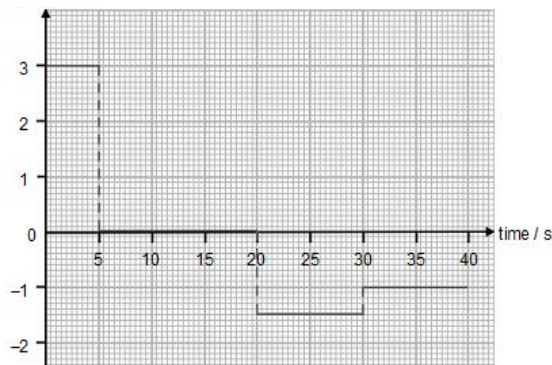


23. The velocity–time graph of a car travelling along a straight line is shown below.

- Draw the corresponding acceleration–time graph of the car.
- When will the car be furthest from the starting point?
- Find the distance travelled by the car during the first 40 s.



Ans:(a) Acceleration/ ms^{-2}

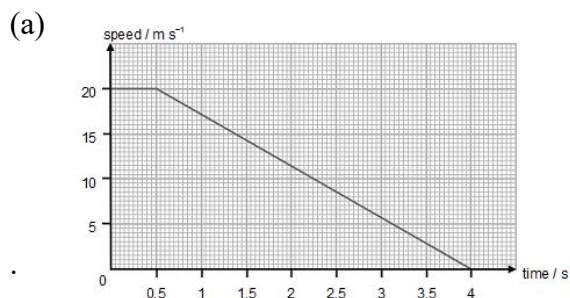


- The car is furthest from the starting point at $t = 30$ s.
- Distance travelled by the car

$$\begin{aligned}
 &= \text{area under the graph} \\
 &= \frac{(15 + 30) \times 15}{2} + \frac{10 \times 10}{2} \\
 &= 388 \text{ m}
 \end{aligned}$$

24. A car is travelling at a uniform speed of 20 m s^{-1} . At time $t = 0$ s, a traffic lights 40 m in front of the car turns red. After a reaction time of 0.5 s, the driver applies the brakes to bring the car to a stop at $t = 4$ s with uniform deceleration.

- Draw the speed–time graph of the car from $t = 0$ s to $t = 4$ s.
- Find the area under the graph and state its physical meaning.
 - Does the car stop in front of or beyond the traffic light? Explain briefly.



(b) (i) Area under the graph = $\frac{(0.5 + 4) \times 20}{2} = 45 \text{ m}$

(ii) This area is equal to the stopping distance of the car.

As the stopping distance is greater than 40 m, the car stops beyond the traffic light.

25. A stone is thrown vertically downwards at an initial velocity of 5 m s^{-1} from the top of a cliff at time $t = 0$. The stone reaches the sea at $t = 2 \text{ s}$. Take the acceleration due to gravity as 9.81 m s^{-2} . Neglect air resistance.
- (a) Find the height of the cliff. (2 marks)
- (b) Find the velocity of the stone when it reaches the sea. (2 marks)
- (c) Sketch the $s-t$, $v-t$ and $a-t$ graphs of the stone from $t = 0$ to $t = 2 \text{ s}$. (3 marks)

Take downwards as positive.

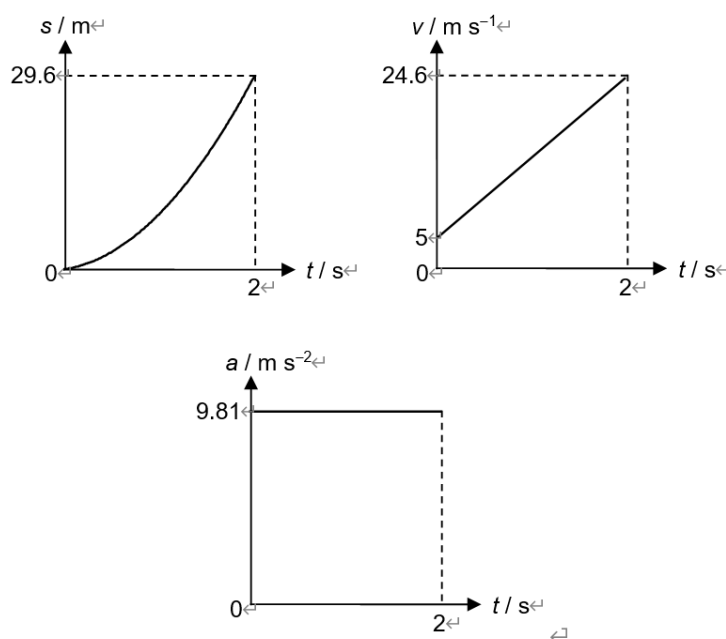
(a) By $s = ut + \frac{1}{2}at^2$,

$$\text{height of cliff} = 5 \times 2 + \frac{1}{2} \times 9.81 \times 2^2 = 29.6 \text{ m}$$

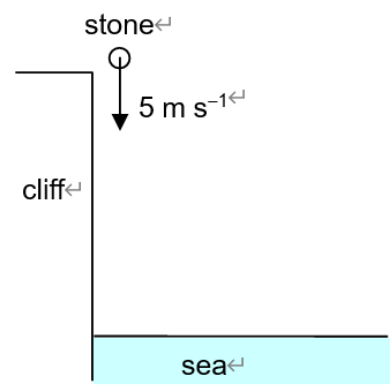
(b) By $v = u + at$,

$$\text{velocity of the stone} = 5 + 9.81 \times 2 = 24.6 \text{ m s}^{-1}$$

(c)



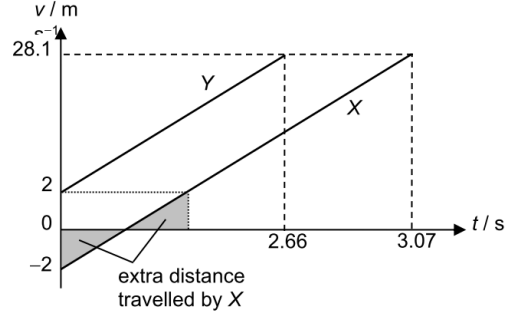
(Each correct graph)

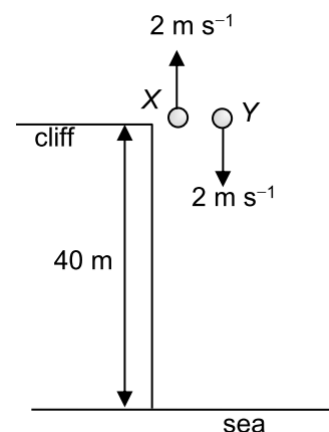


26. A stone X is thrown vertically upwards at 2 m s^{-1} from the edge of a cliff which is 40 m above the sea. At the same time, a second stone Y is thrown vertically downwards at 2 m s^{-1} from the same position. Neglect air resistance.

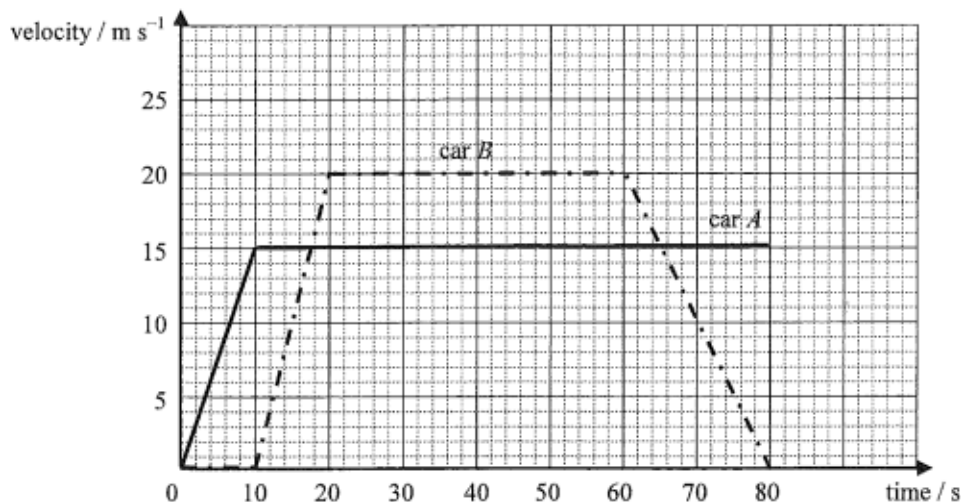
Take downwards as positive.

- Find the velocities of X and Y when they reach the sea. (3 marks)
- When do X and Y reach the sea? (3 marks)
- Sketch in the same graph the v - t graphs for X and Y when they stay in air. (3 marks)
- In the graph in (c), shade the region that represents the extra distance travelled by X . (1 mark)

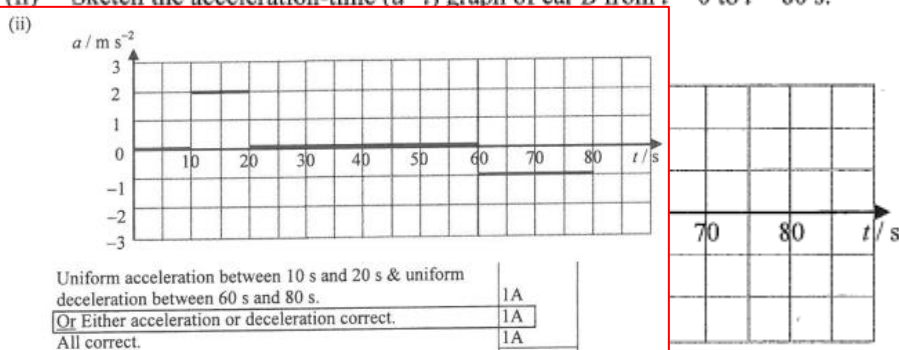
(a) Apply $v^2 = u^2 + 2as$.	1M
Velocity of $X = \sqrt{(-2)^2 + 2 \times 9.81 \times 40} = 28.1 \text{ m s}^{-1}$	1A
Velocity of $Y = \sqrt{2^2 + 2 \times 9.81 \times 40} = 28.1 \text{ m s}^{-1}$	1A
(b) Apply $v = u + at$.	1M
Time for X to reach the sea $= \frac{28.1 - (-2)}{9.81} = 3.07 \text{ s}$	1A
Time for Y to reach the sea $= \frac{28.1 - 2}{9.81} = 2.66 \text{ s}$	1A
(c)	
	
(Correct labelled axes with units)	1A
(Correct straight line for X)	1A
(Correct straight line for Y)	1A
(d) (Correct shaded area)	1A



27. Two cars A and B initially at the same position, start to travel along the same straight horizontal road. The graph below shows how their velocities vary with time.



- (a) Describe the motion of car A along the whole journey from $t = 0$ to $t = 80$ s.
(b) (i) Which car attained the greatest acceleration throughout the journey? Find this acceleration.
(ii) Sketch the acceleration-time ($a-t$) graph of car B from $t = 0$ to $t = 80$ s.



- (c) (i) At $t = 20$ s, what is the separation between cars A and B ?
(ii) Deduce the time at which car B catches up with car A .

- (a) 0 – 10 s: constant / uniform acceleration 1A
10 – 80 s: constant velocity / uniform motion 1A

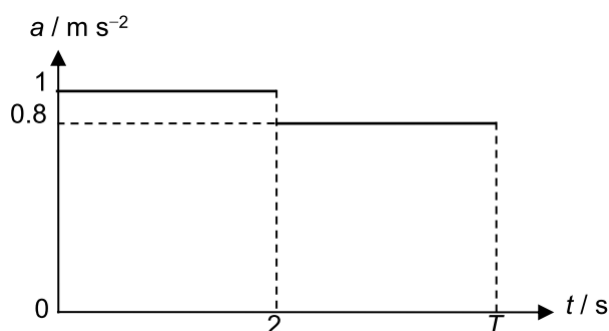
- (b) (i) Car B . 1A
It is represented by the steepest slope between 10 s and 20 s for B .
 $a = \frac{20 - 0}{20 - 10} = 2 \text{ m s}^{-2}$ 1A

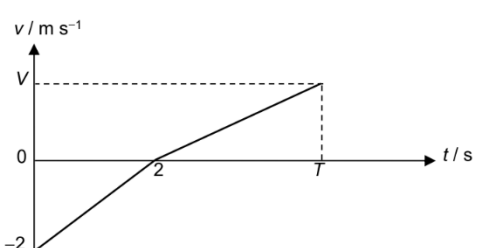
- (c) (i) Total area of A by 20 s = $\frac{(10 + 20)}{2} \times 15 = 225 \text{ m}$ 1M
Total area of B by 20 s = $\frac{(10 \times 20)}{2} = 100 \text{ m}$
Separation of A & B , $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$ 1A

- (ii) Let B meet A at time $(20 + T)$ s, 1M
 $(v_B - v_A) \times T = s_{AB}$
 $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$
 $T = 25 \text{ s}$
 $\therefore B$ catches up with A at $t = (20 + 25) \text{ s} = 45 \text{ s}$. 1A
Or $s_A + v_A \times T = s_B + v_B \times T$ 1M
 $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$
Or Accept using graphical method and read the answer from the graph 1M
1A

28. At time $t = 0$, an object is given a sharp push such that it moves up an inclined plane. The object reaches the highest position and is momentarily at rest at $t = 2$ s. Then it slides down the plane and returns to the starting position at time T . The figure below shows its $a-t$ graph.

- (a) Which direction is taken as positive in the $a-t$ graph? (1 mark)
 (b) Describe the motion of the object. (2 marks)
 (c) Find the initial velocity of the object. (2 marks)
 (d) Sketch the $v-t$ graph of the object. It is **not** necessary to calculate the final velocity V of the object. (3 marks)
 (e) Hence, find T . (4 marks)



(a) Down the plane	1A
(b) The object moves up the plane and decelerates to rest uniformly from 0–2 s. It accelerates uniformly from rest down the plane from 2 s to T .	1A 1A
(c) By $v = u + at$, $0 = u + 1 \times 2$ $u = -2 \text{ m s}^{-1}$ The initial velocity is 2 m s^{-1} up the plane.	1M 1A
(d) 	1A 1A 1A
(e) Displacement = area under $v-t$ graph	
$0 = \frac{1}{2} [(-2) \times 2 + V(T - 2)]$	1M
$VT - 2V - 4 = 0 \dots\dots\dots(1)$ Acceleration = slope of $v-t$ graph $0.8 = \frac{V - 0}{T - 2}$	1M
$V = 0.8T - 1.6 \dots\dots\dots(2)$ Substituting (2) into (1), $(0.8T - 1.6)T - 2(0.8T - 1.6) - 4 = 0$	1M
$\Rightarrow T^2 - 4T - 1 = 0$ $\Rightarrow T = 4.24 \text{ s}$ or -0.236 s (rejected)	1A